

Final Project Report: [Project Name]

Team Members: [Names]

Course: [Course Name]

Date: [Submission Date]

1. Introduction

- **Problem Definition:** Reiterate the everyday problem and its significance.
- **Objective:** Clearly state what the application achieves.
- **Project Scope:** Briefly outline the features included in the final prototype.

2. Related Work

- Briefly discuss any existing solutions to this problem and how your AI-based approach differs or improves upon them.

3. Data Acquisition and Preprocessing

- **Dataset Description:** Source, size, features, and target labels.
- **Data Preprocessing Pipeline:**
 - Cleaning (handling missing values, outliers).
 - Augmentation techniques (if applicable).
 - Normalization/Vectorization methods.
- **Data Split:** Details on training, validation, and testing sets.

4. Methodology & Technical Approach

- **System Architecture:** A high-level diagram showing how the frontend, backend, and AI model interact.
- **Model Selection:** Justification for the chosen algorithm(s) (e.g., CNN, RNN, Random Forest).
- **Training Process:**
 - Hyperparameters used (learning rate, epochs, batch size, etc.).
 - Training environment (GPU/CPU used).
 - Techniques to avoid overfitting (dropout, regularization, early stopping).

5. Evaluation and Results

- **Metrics:** Definition of metrics used (Accuracy, Precision, Recall, F1, MSE, etc.).

- **Results:** Present findings using tables and graphs (e.g., Confusion Matrix, ROC curve, Loss vs. Epoch plot).
- **Discussion:** Analyze the performance. Did the model meet the latency requirements (< 2s)?

6. User Interface and Integration

- **UI Design:** Screenshots of the application and description of user flow.
- **Integration:** How the model was deployed within the app (e.g., Flask API, TensorFlow.js, etc.).

7. Ethical Considerations and Limitations

- **Limitations:** What can the model *not* do? What are its failure points?
- **Ethical Considerations:** Discussion on data privacy, potential biases in the training data, and environmental impact of training.

8. Conclusion and Future Work

- Summary of achievements.
- Potential improvements for future iterations (e.g., deploying to cloud, increasing dataset size, improving UI).

9. References

- Cite all datasets, libraries, pre-trained models, and academic papers used.

Final Presentation & Demo Script Template

Time Limit: 10 Minutes

1. Introduction (1 Minute)

- **Hook:** Briefly state the everyday problem (e.g., "Have you ever thrown away food because you forgot it was in the back of the fridge?")
- **The Solution:** Introduce your app name and its core purpose.
- **Value Proposition:** What is the primary benefit to the user?

2. The Problem & User Need (1 Minute)

- Describe the target user persona.
- Explain why existing solutions (non-AI) are inadequate.

3. Technical Approach: AI Model (3 Minutes)

- **Data:** Briefly describe the dataset used to train the model.
- **Architecture:** Explain *why* you chose this specific AI approach (CV, NLP, ML) and model architecture.
- **Training:** Mention any interesting challenges faced during training (e.g., overfitting, data imbalance) and how you solved them.

4. Live Demonstration (3 Minutes)

- **Scenario:** Walk through a realistic user scenario.
- **Action:** Show the user inputting data (uploading photo, speaking, typing).
- **Reaction:** Show the AI processing the input and displaying the result.
- **Speed:** Highlight that the inference time meets requirements (< 2s).

5. Evaluation & Results (1 Minute)

- **Metrics:** State the final accuracy/performance metrics (F1-score, MSE, etc.).
- **Validation:** Briefly show a confusion matrix or key graph to prove model reliability.

6. Conclusion & Future Work (1 Minute)

- **Summary:** Reiterate how the app solves the everyday problem.
- **Limitations:** Honestly mention what the model cannot do well.
- **Future Scope:** What is one feature you would add next?
- **Q&A:** Open floor for questions.